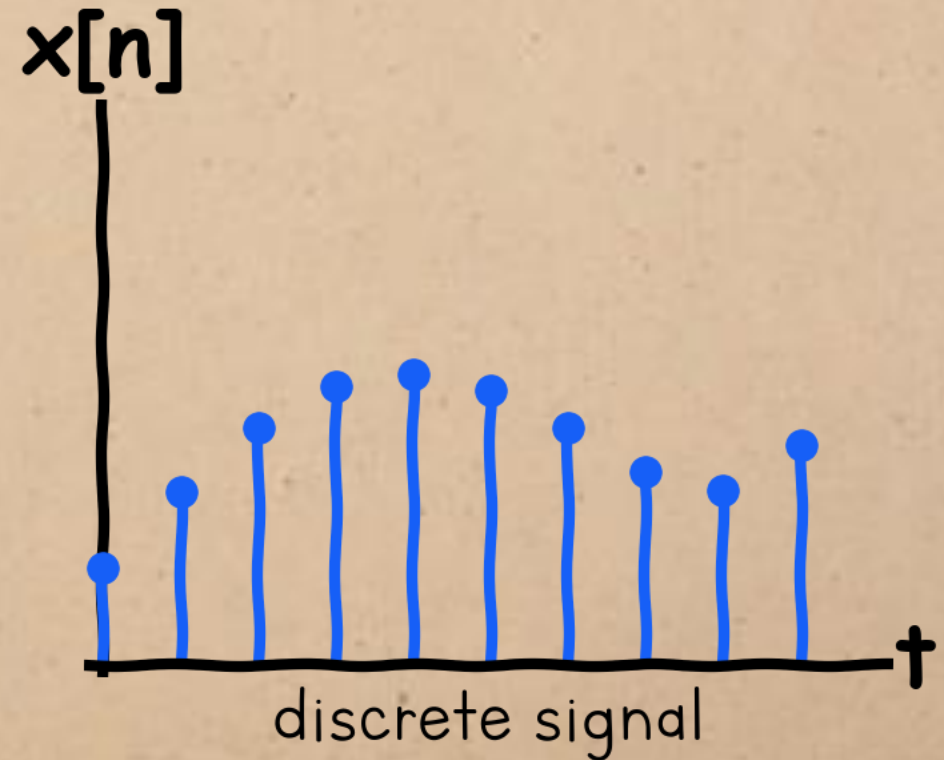
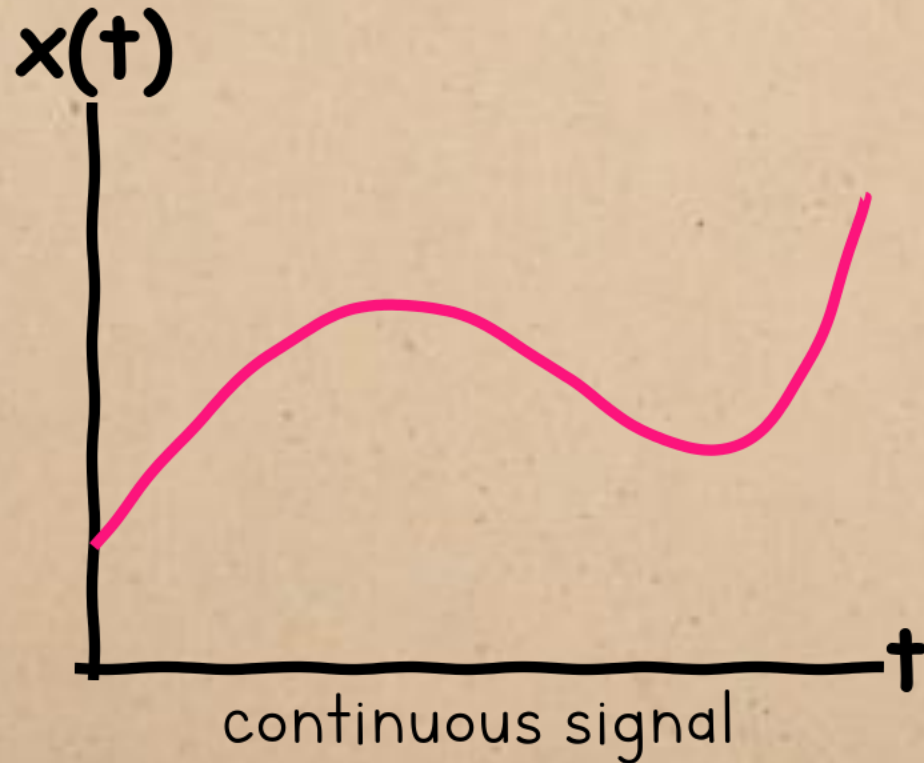




SIGNALS & SYSTEMS RECAP: IMPULSE RESPONSE

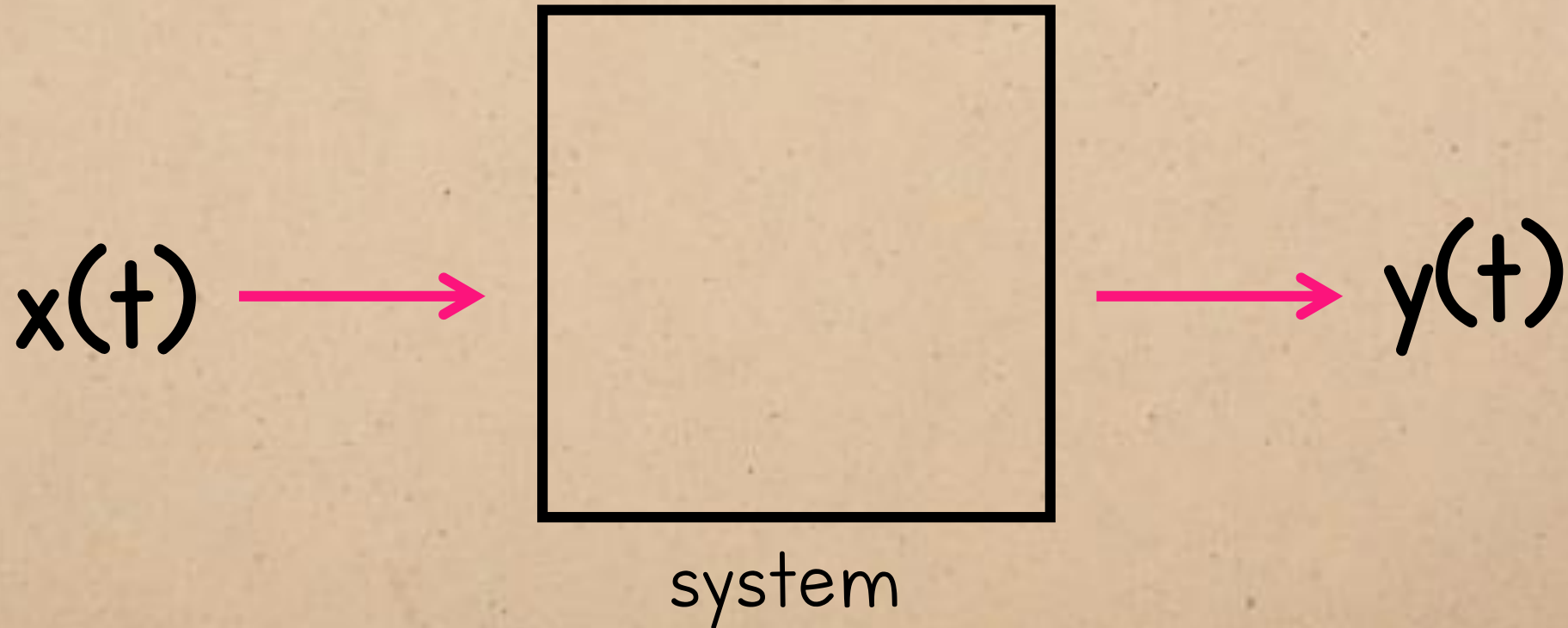
SIGNALS

A signal is a quantity that varies with time.



SYSTEMS

Systems are objects or processes that **transform** signals.



SYSTEMS

Anything can be a system, if you can think of suitable inputs and outputs to describe it that way.



car



motor



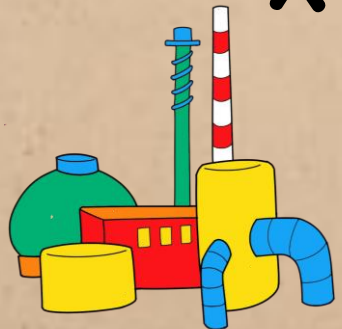
resistor



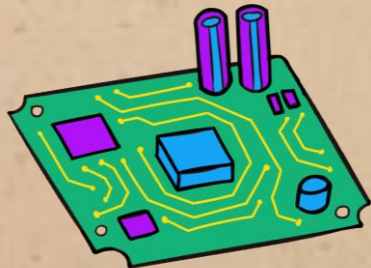
chihuahua



robot



chemical plant



circuit



quadcopter



guy covered in bees



avocado

$x(t)$



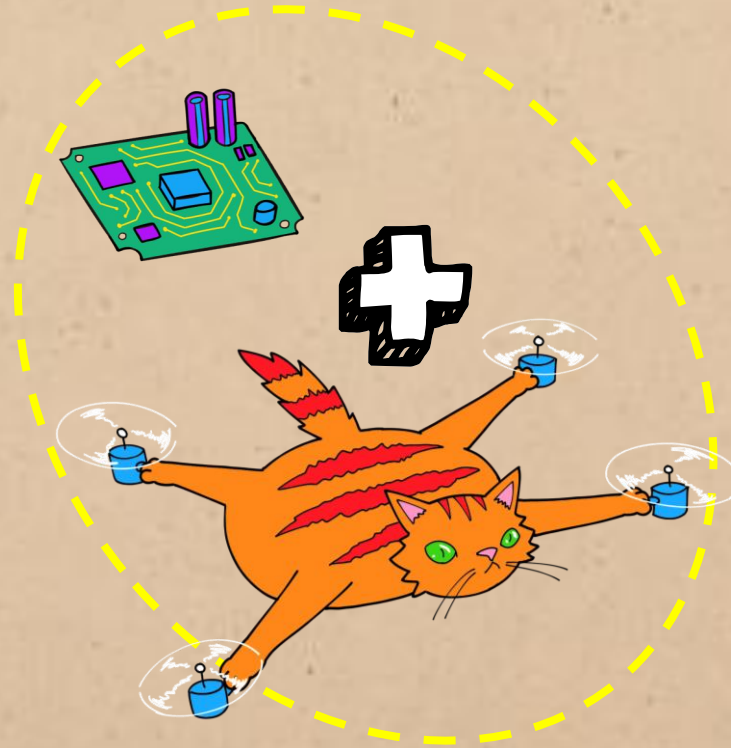
$y(t)$

SYSTEMS

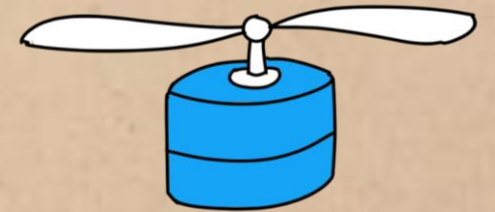
In control engineering, “the system” could refer to:



the thing we're trying
to control



the combination of the
thing and the controller



part of the thing

SYSTEMS

A system is an **input-output** relationship between signals.



SYSTEMS

Multiple-input, multiple-output (MIMO) system:



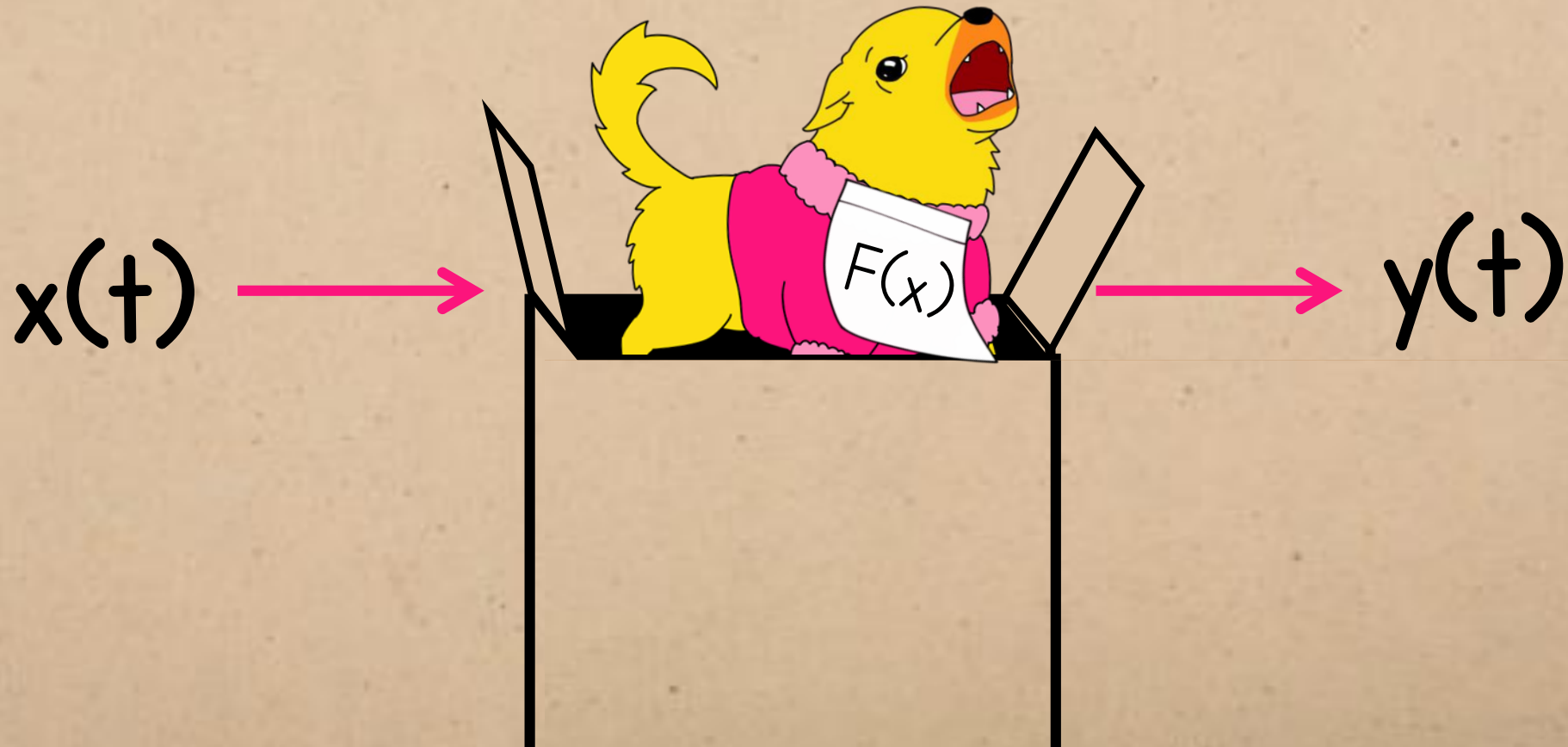
SYSTEMS

Single-input, single-output (SISO) system:



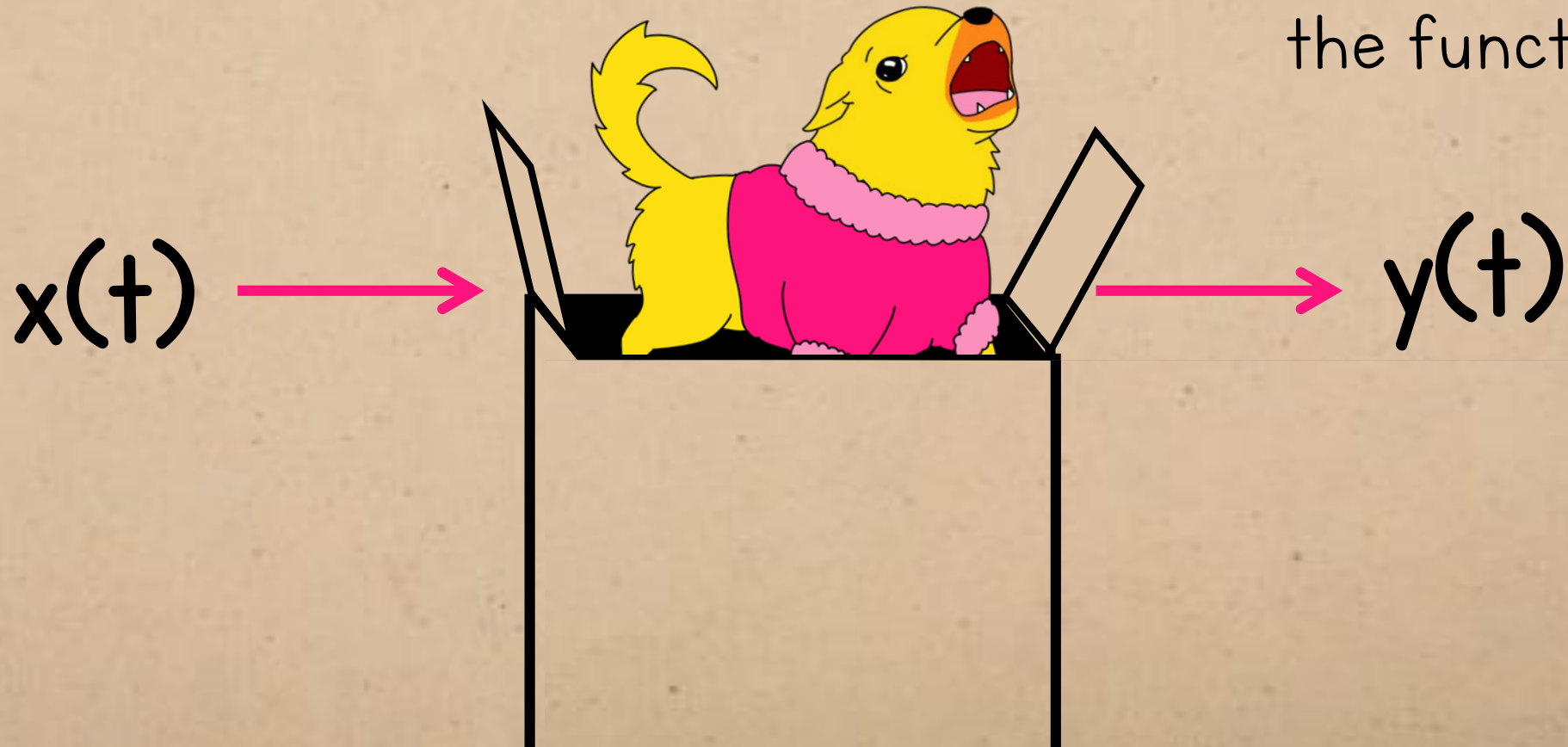
SYSTEMS

We can think of a signal as a mathematical function.



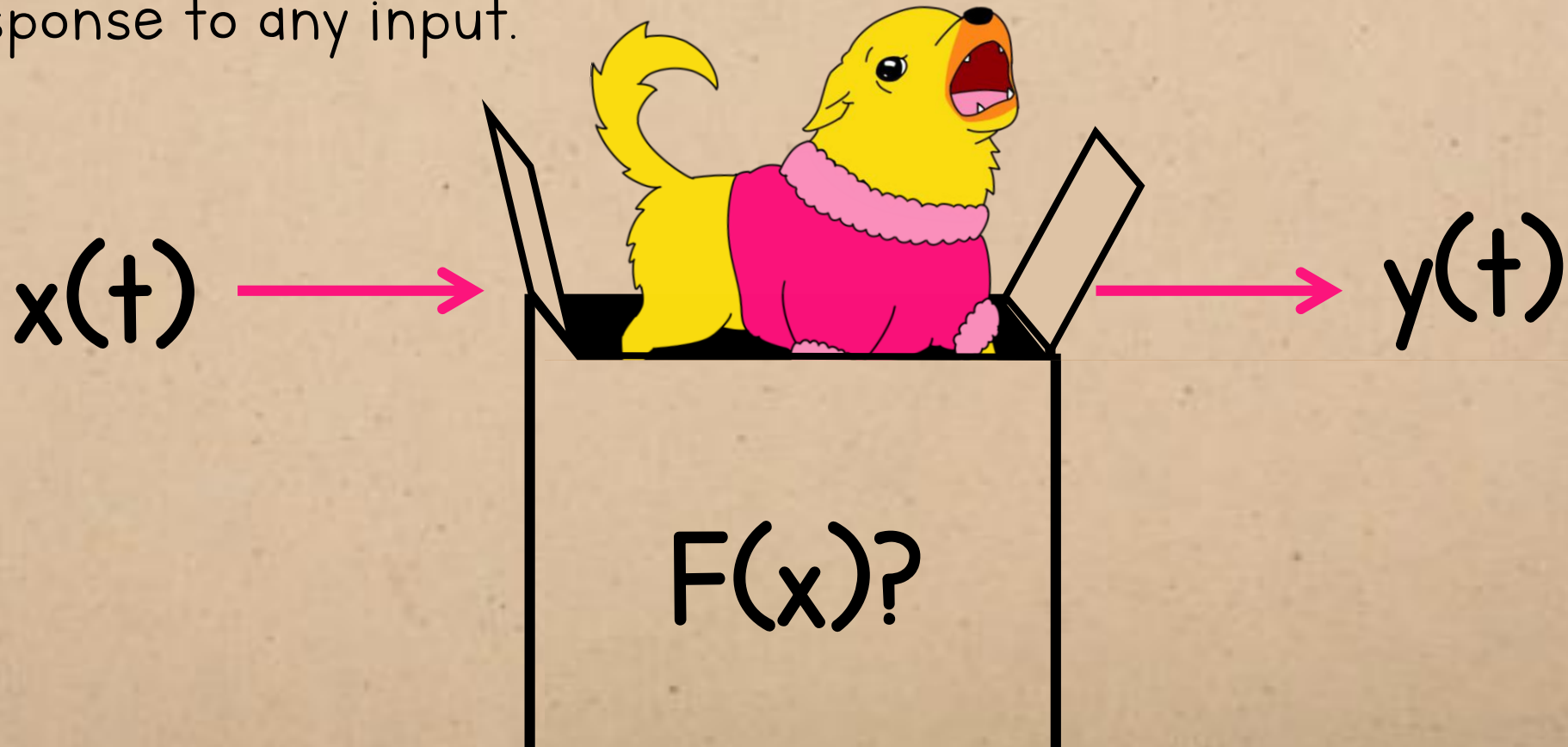
SYSTEMS

...But we won't
always know what
the function is.



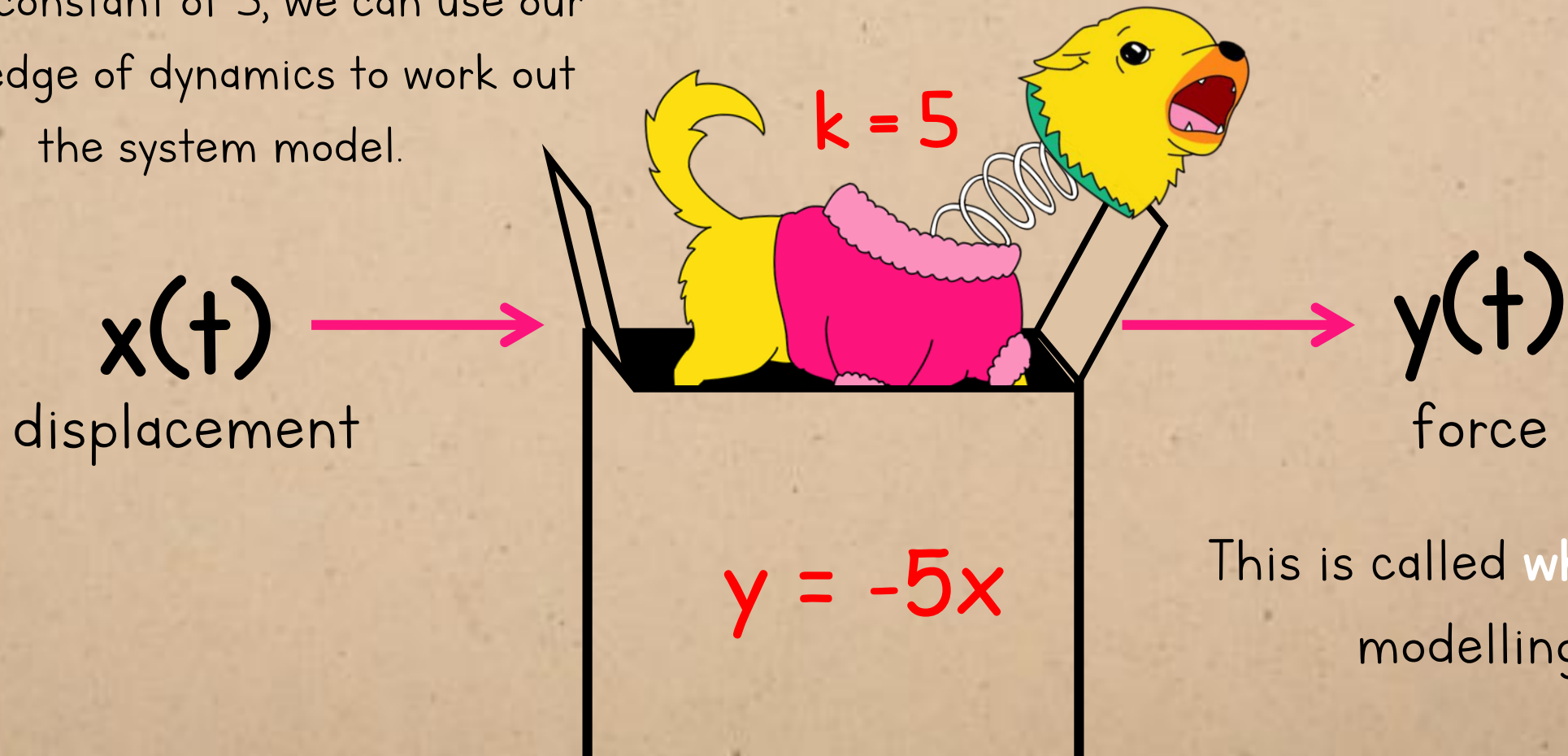
SYSTEM ID.

System identification = finding
a model that can calculate the
response to any input.



SYSTEM ID.

If we know something about the system, e.g. that it contains a spring with a constant of 5, we can use our knowledge of dynamics to work out the system model.

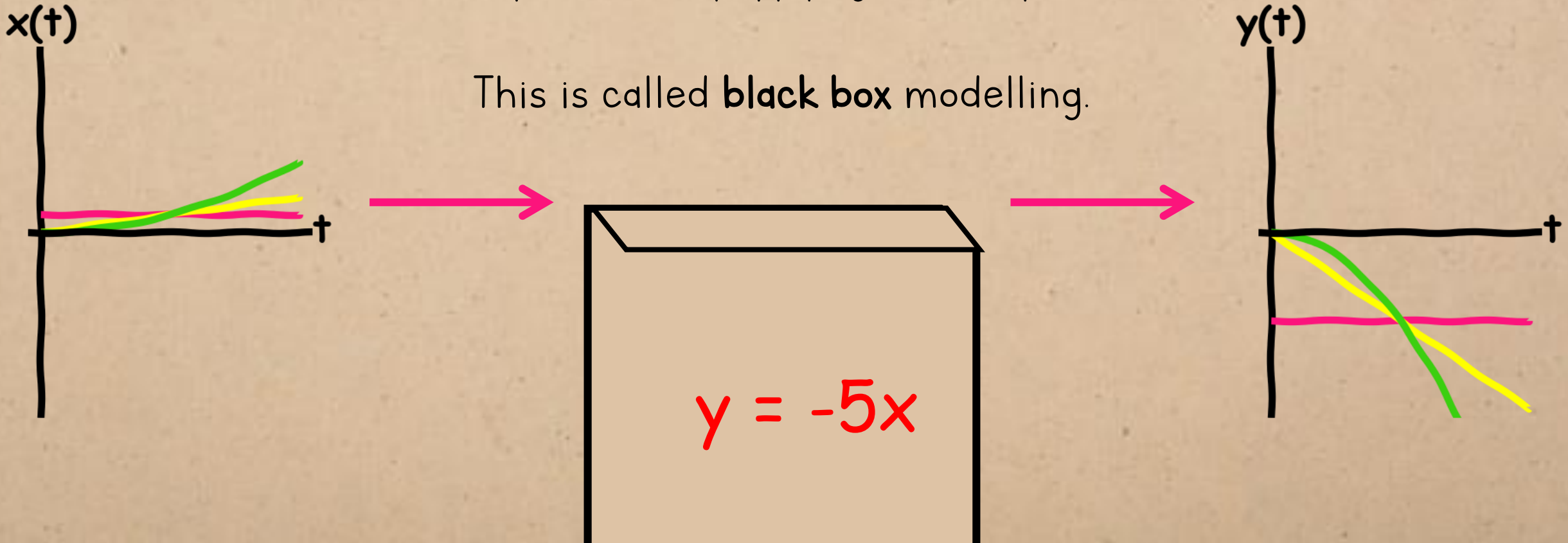


This is called **white box** modelling.

SYSTEM ID.

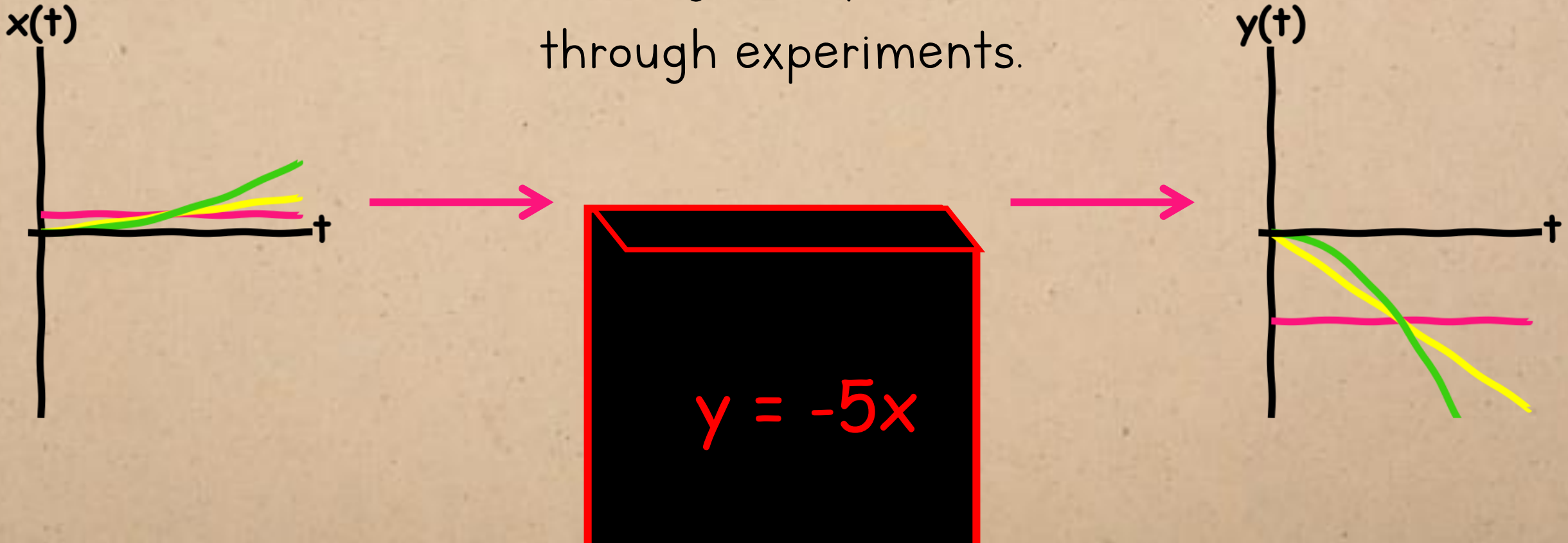
If we don't know anything about the contents of the system, we can identify the model through experiments, by applying known inputs.

This is called **black box** modelling.



SYSTEM ID.

BLACK BOX method =
determining the system model
through experiments.



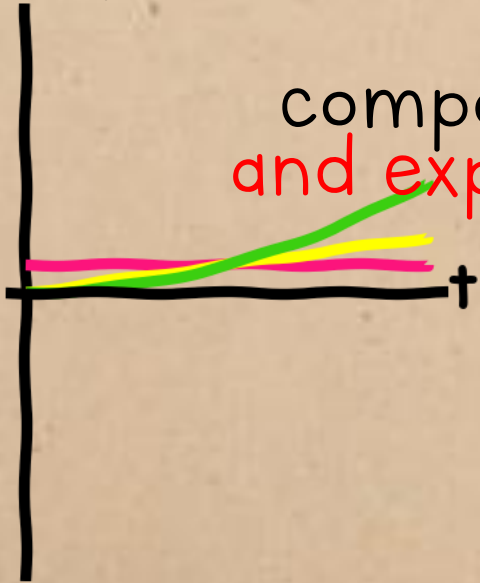
SYSTEM ID.

greyish

~~WHITE BOX~~ method =
determining the system

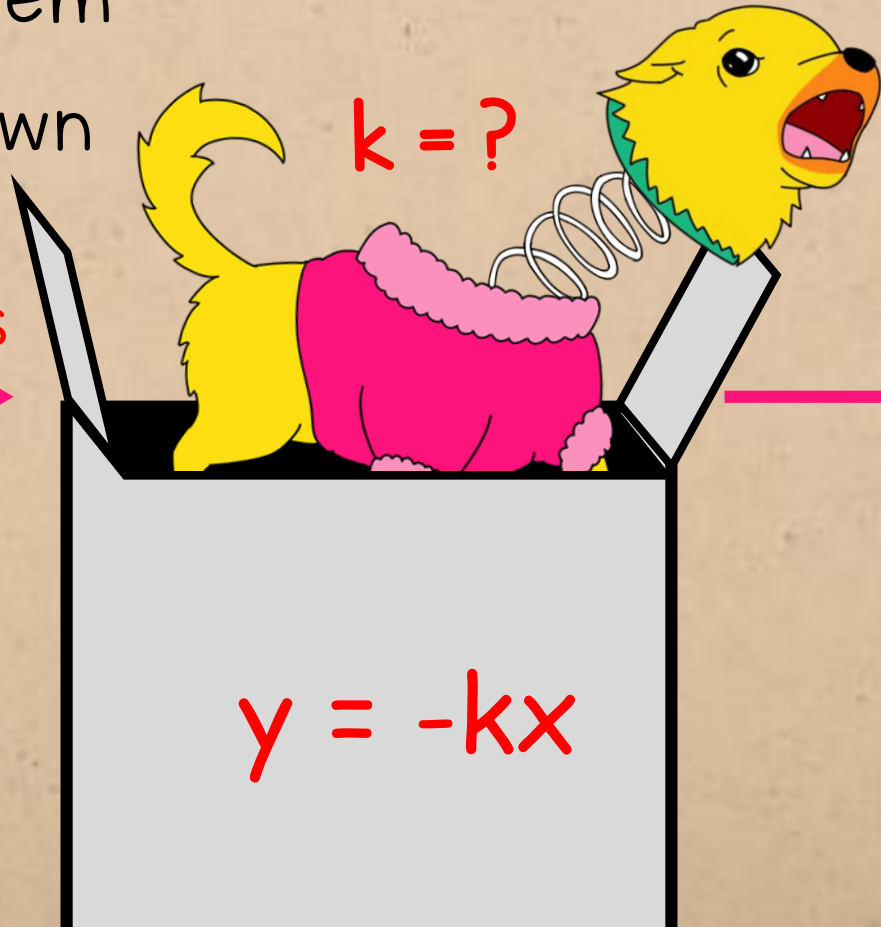
$x(t)$ model based on known

components.
and experiments

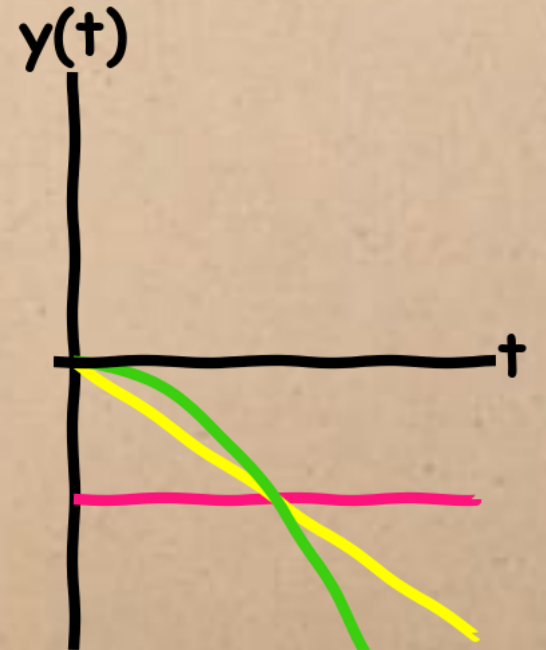


$k = ?$

$$y = -kx$$

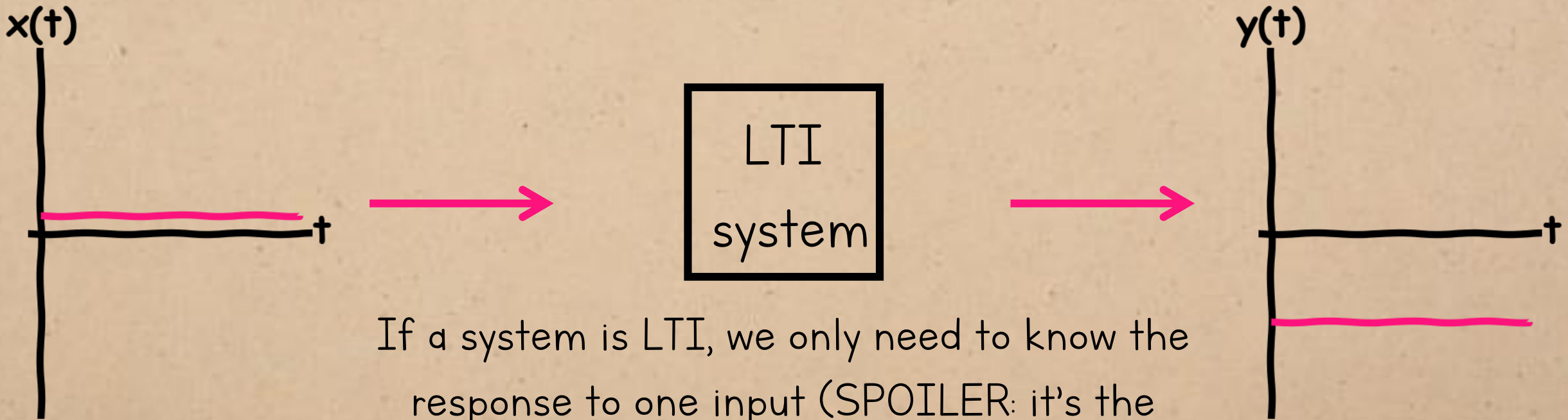


Even when we can work out the general form of the system model, there will probably be some unknown parameters (e.g. the spring constant, here) that we have to find through experiments.



LTI SYSTEMS

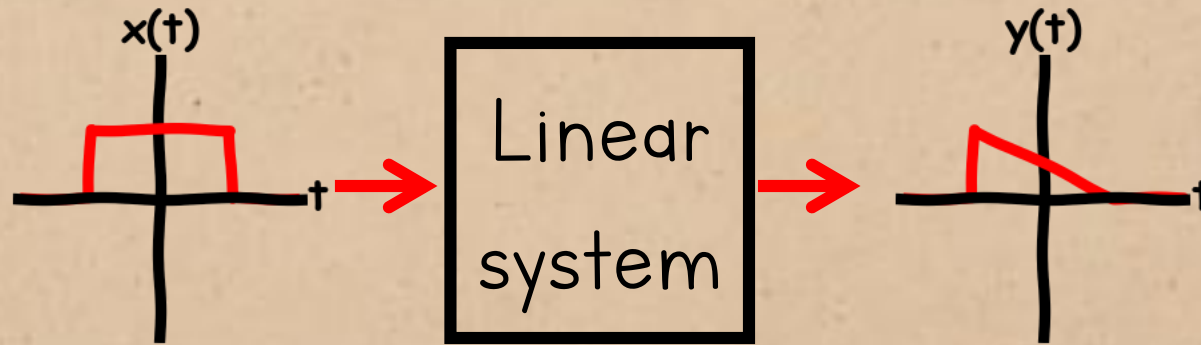
Linear and Time-Invariant (LTI) systems: only need to know the response to one signal to fully define the system.



If a system is LTI, we only need to know the response to one input (SPOILER: it's the impulse) to calculate the response to any input.

LTI SYSTEMS

homogeneity: response to scaled input
= response scaled by same factor

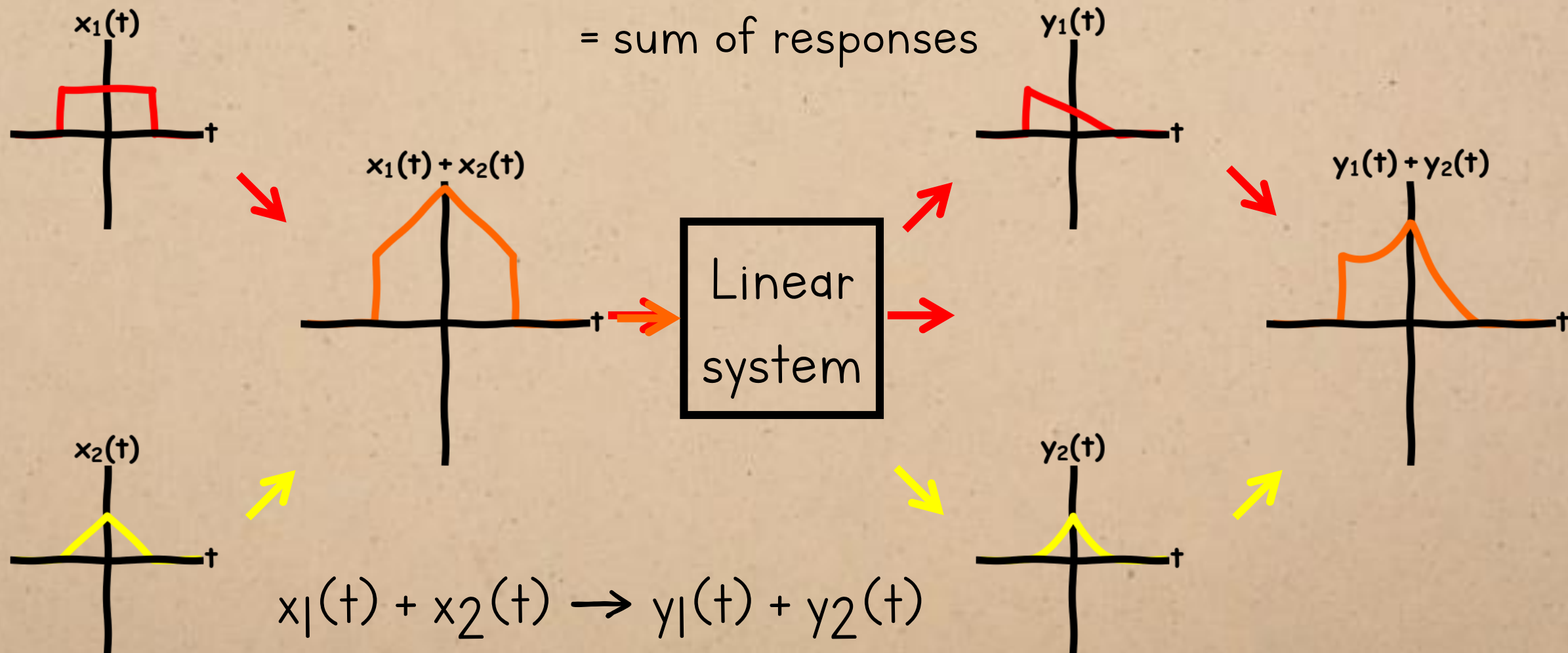


$$ax(t) \rightarrow ay(t)$$

LTI SYSTEMS

Additivity: response to sum

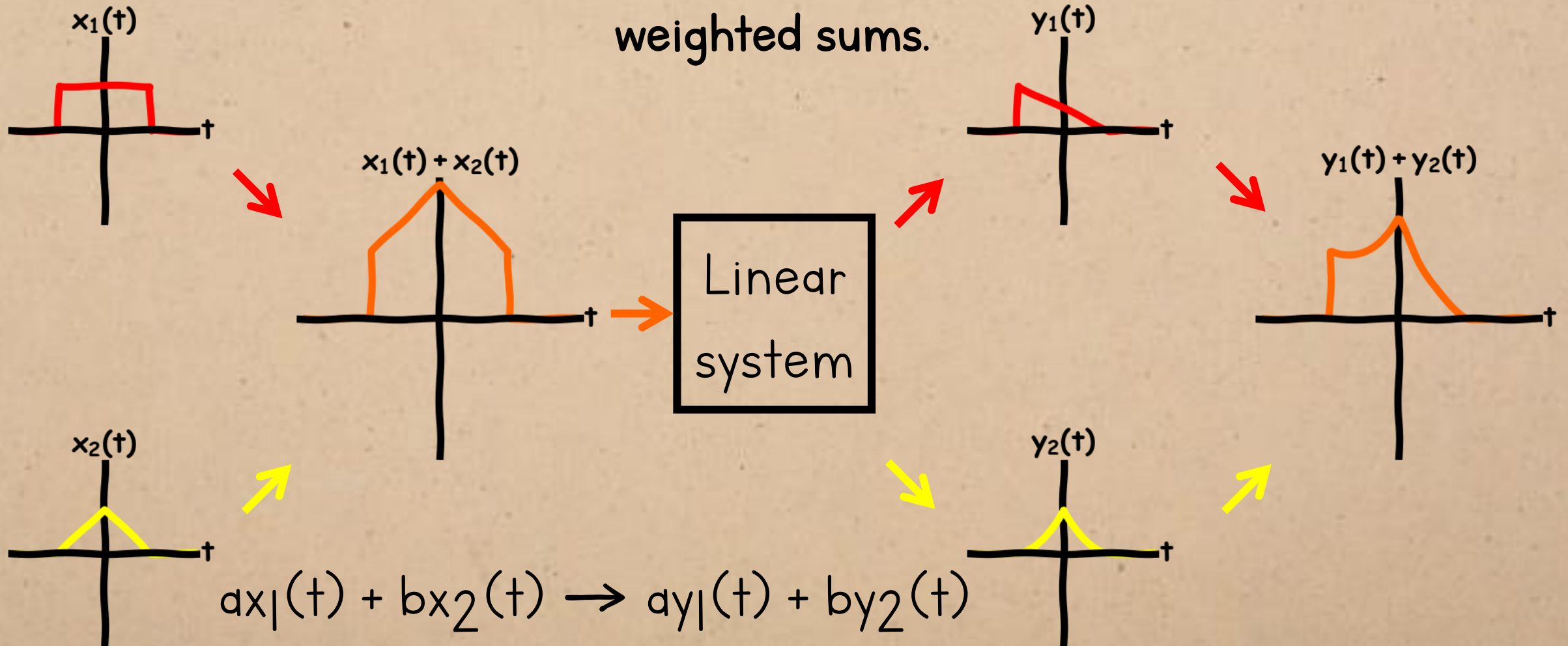
= sum of responses



LTI SYSTEMS

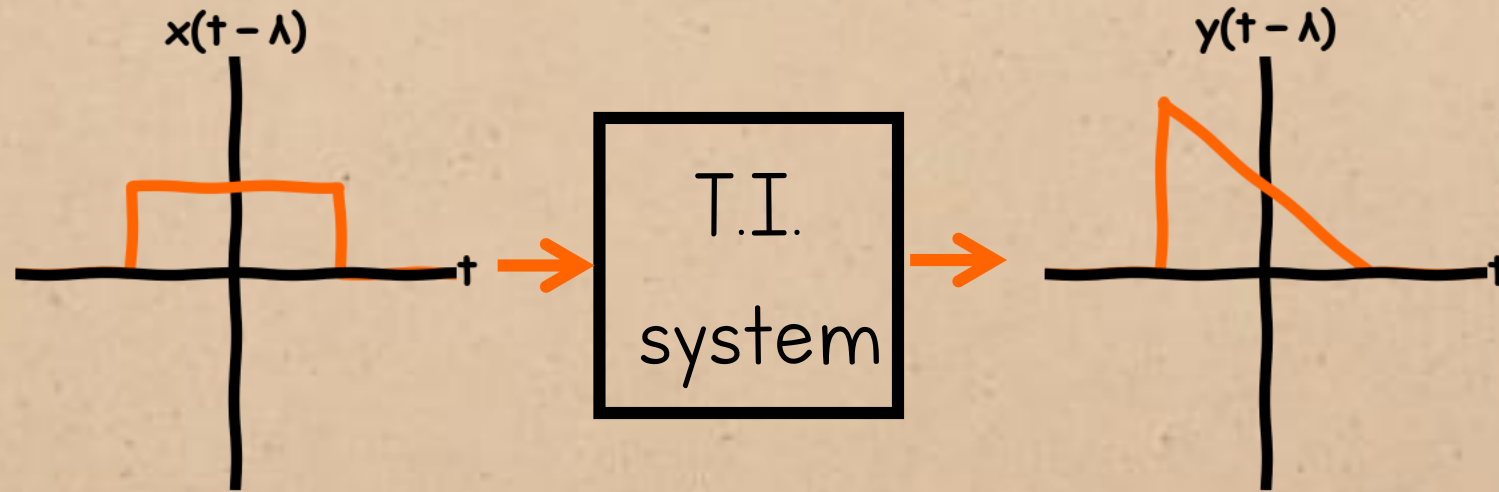
Linear systems preserve

weighted sums.



LTI SYSTEMS

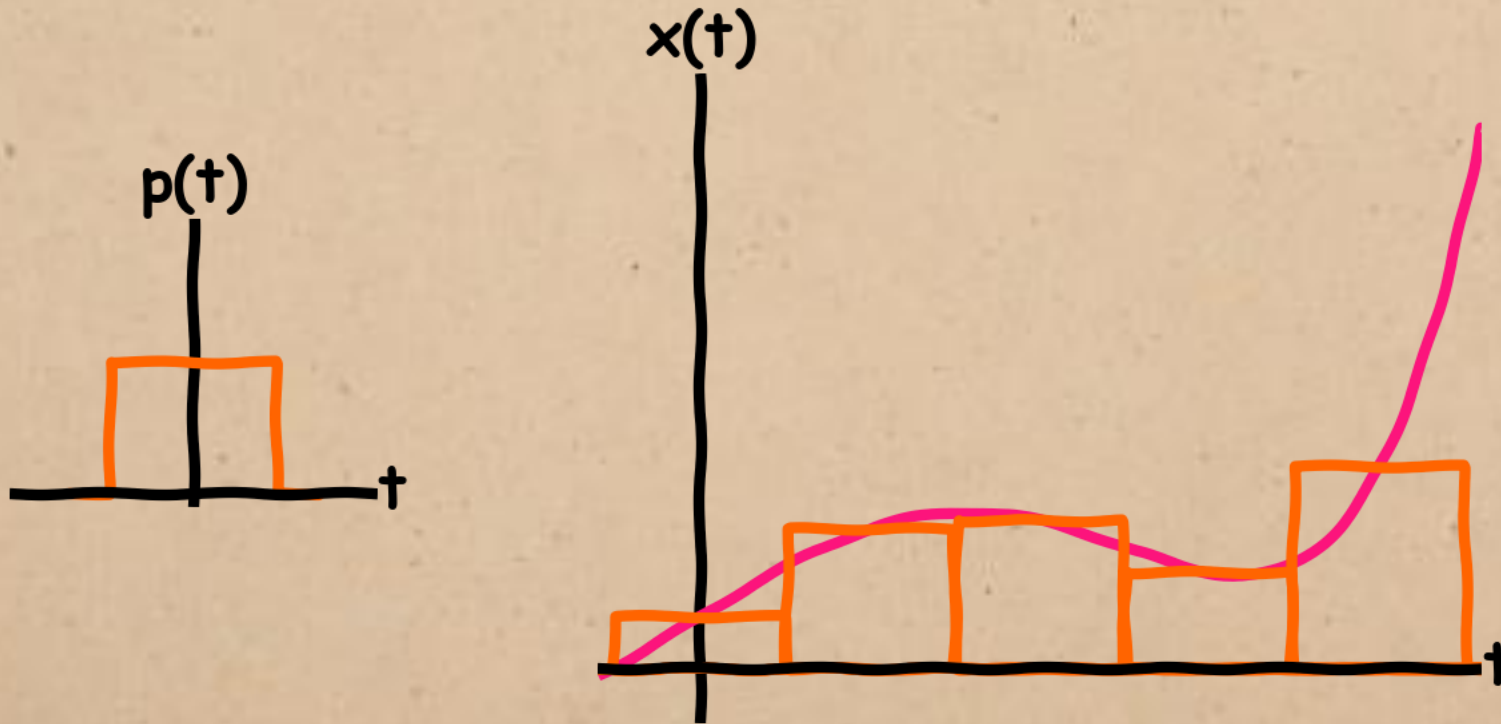
Time invariance: time-shifting the input causes the same shift in the response.



$$x(t - \lambda) \rightarrow y(t - \lambda)$$

LTI SYSTEMS

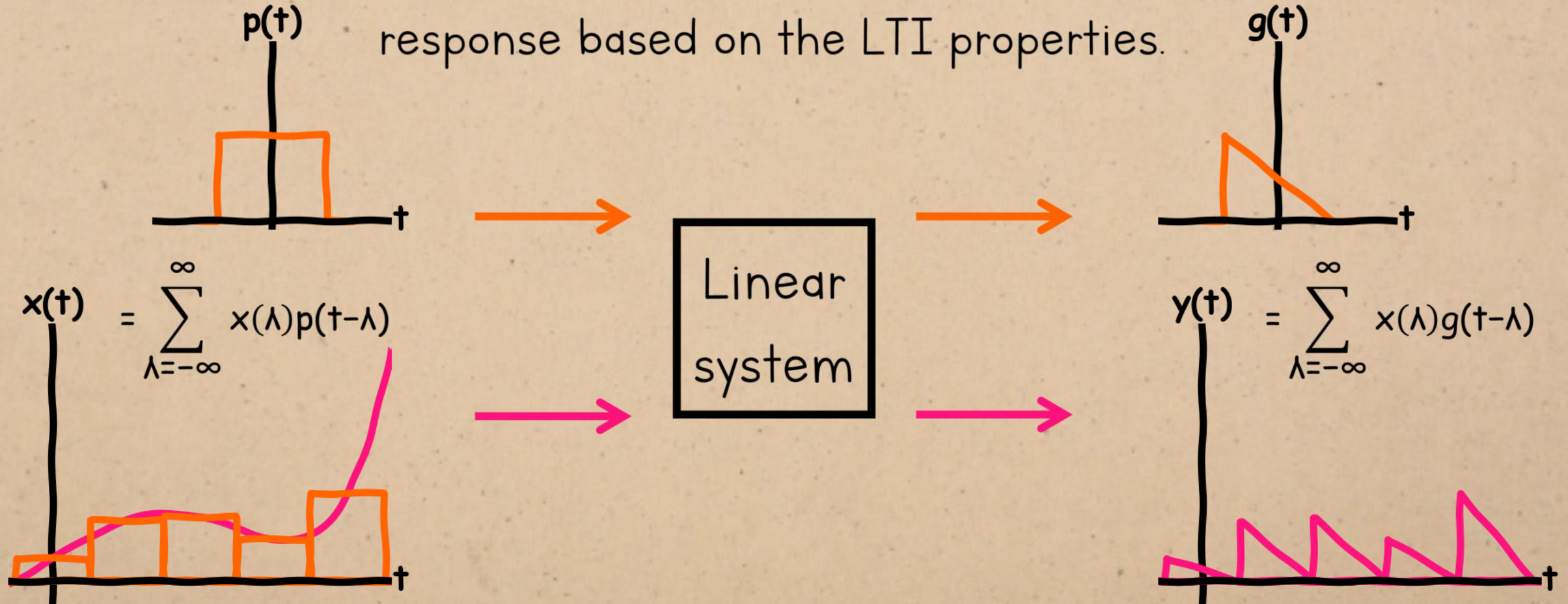
We can approximate any input using a sum of shifted, scaled pulses.



$$= \sum_{\lambda=-\infty}^{\infty} x(\lambda) p(t-\lambda)$$

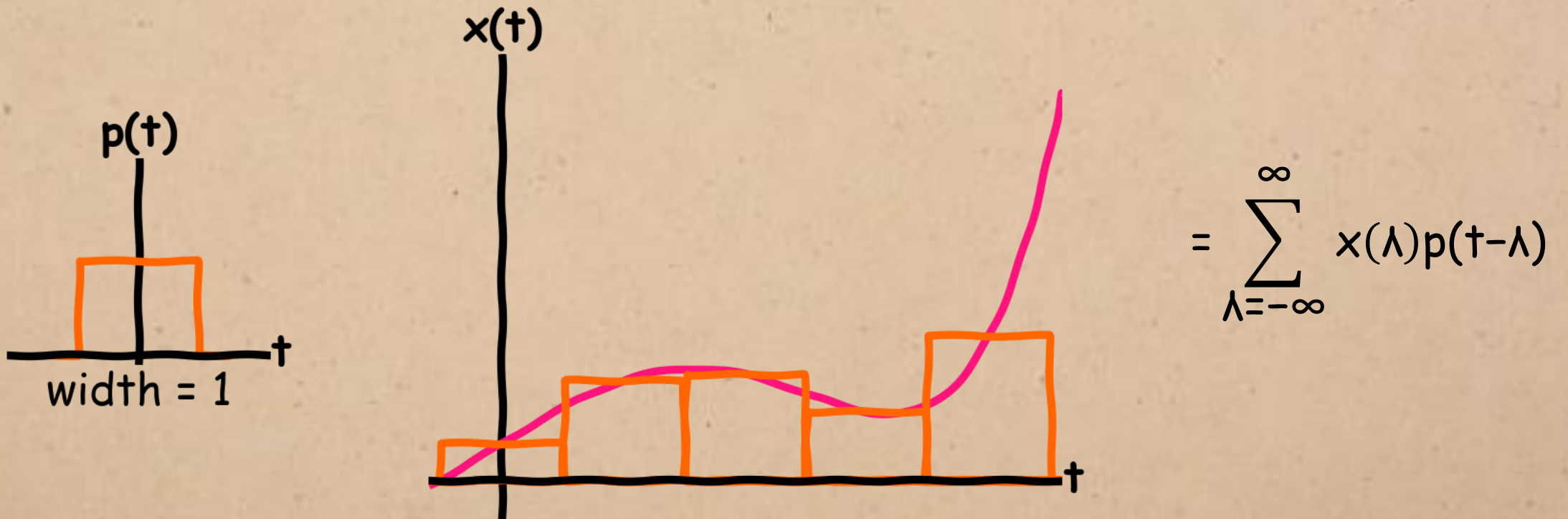
LTI SYSTEMS

...so, if we know the response to the pulse, we can approximate the response based on the LTI properties.



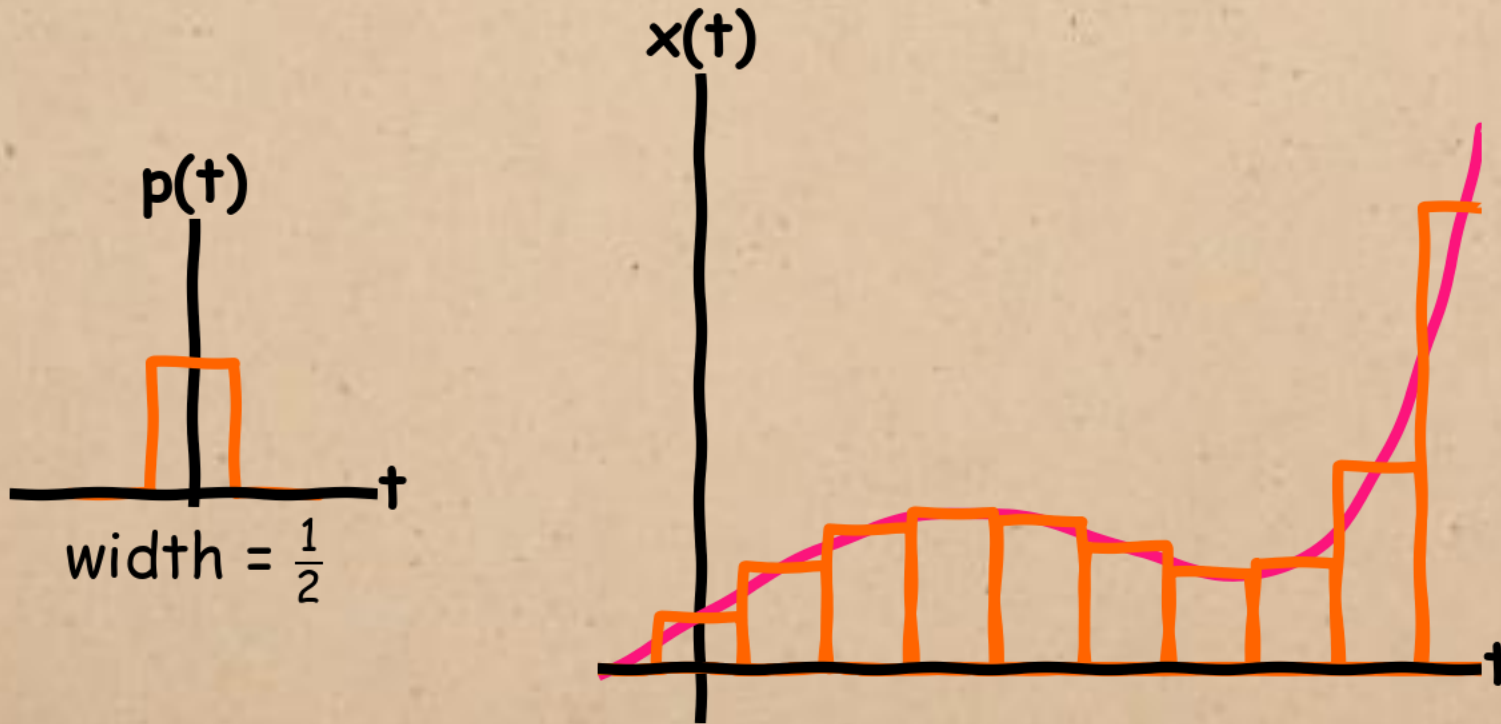
LTI SYSTEMS

Narrower pulses spaced more closely together give a more accurate approximation.



LTI SYSTEMS

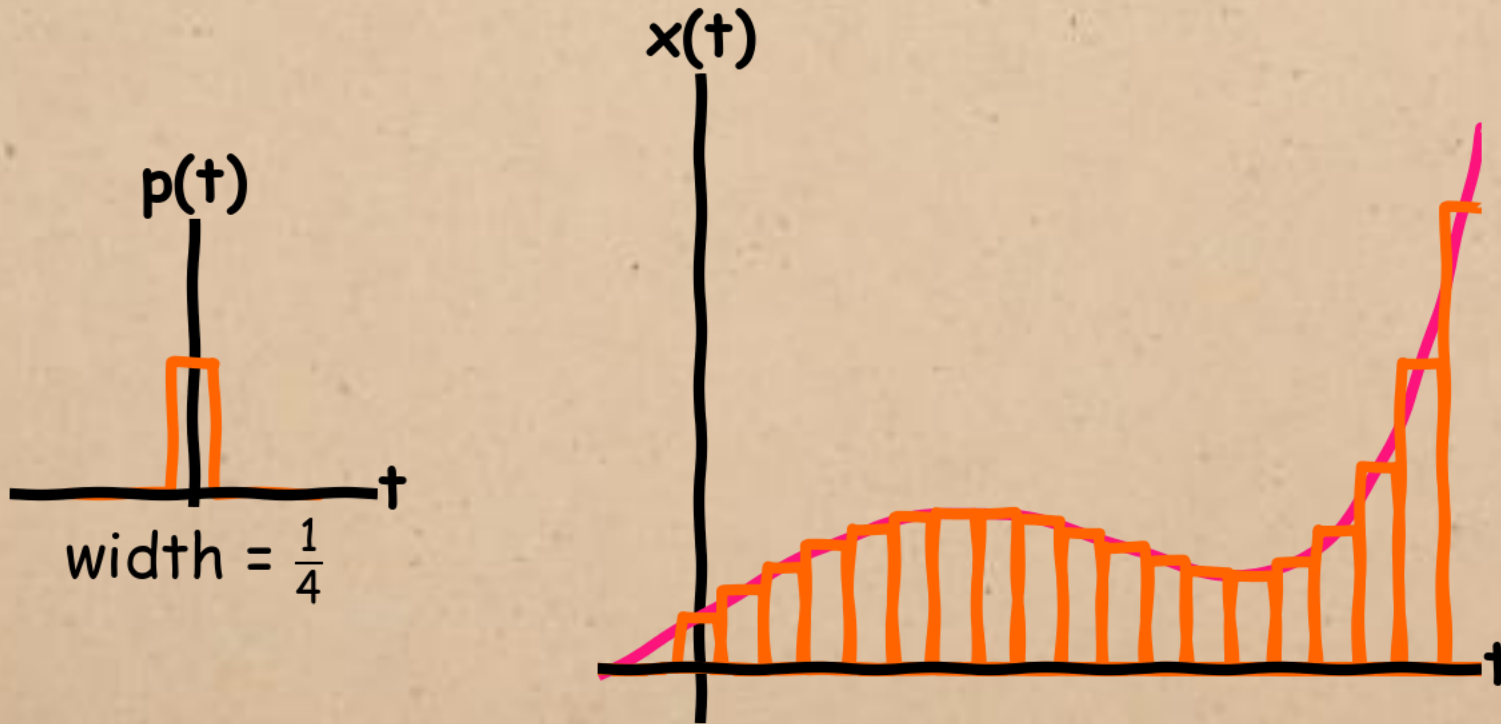
Narrower pulses spaced more closely together give a more accurate approximation.



$$= \sum_{\lambda=-\infty}^{\infty} x\left(\frac{1}{2}\lambda\right) p\left(t - \frac{1}{2}\lambda\right)$$

LTI SYSTEMS

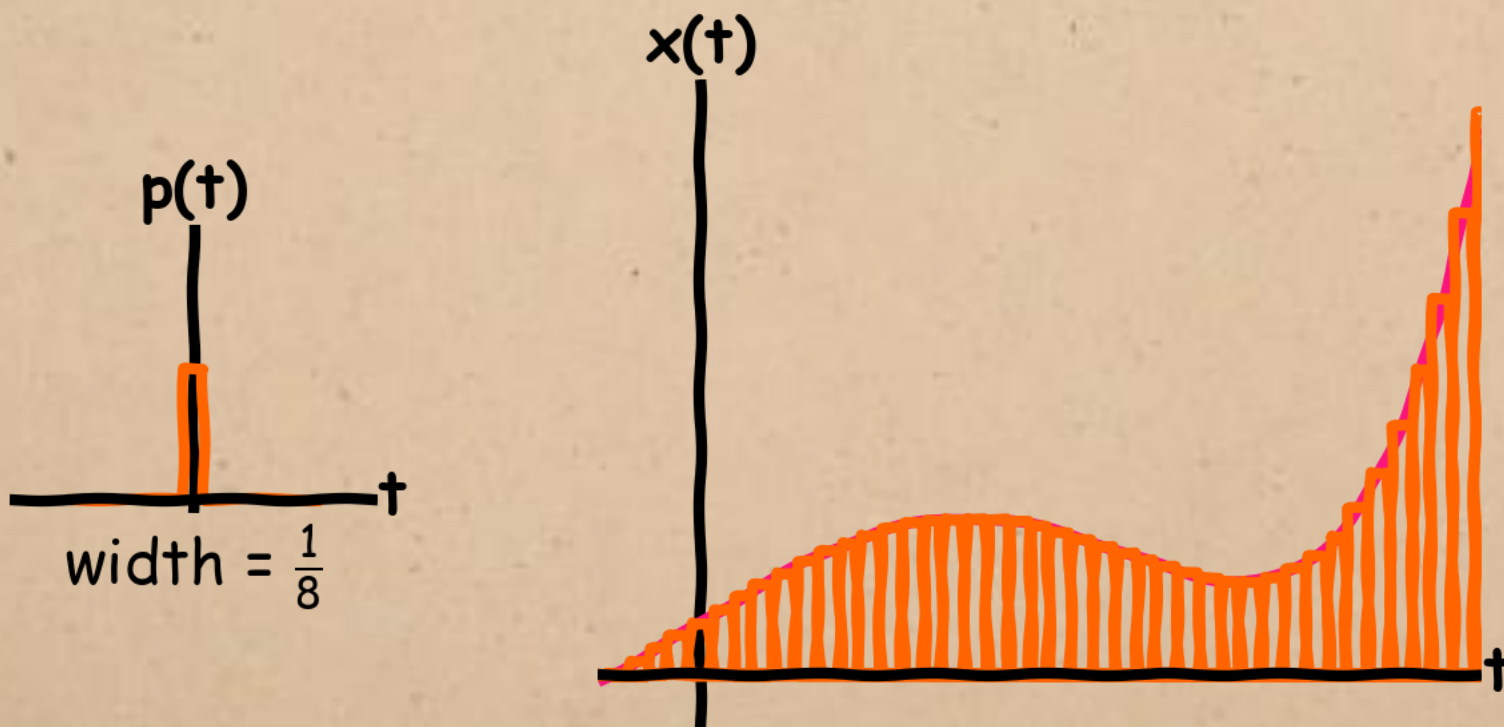
Narrower pulses spaced more closely together give a more accurate approximation.



$$= \sum_{\lambda=-\infty}^{\infty} x\left(\frac{1}{4}\lambda\right) p\left(t - \frac{1}{4}\lambda\right)$$

LTI SYSTEMS

Narrower pulses spaced more closely together give a more accurate approximation.

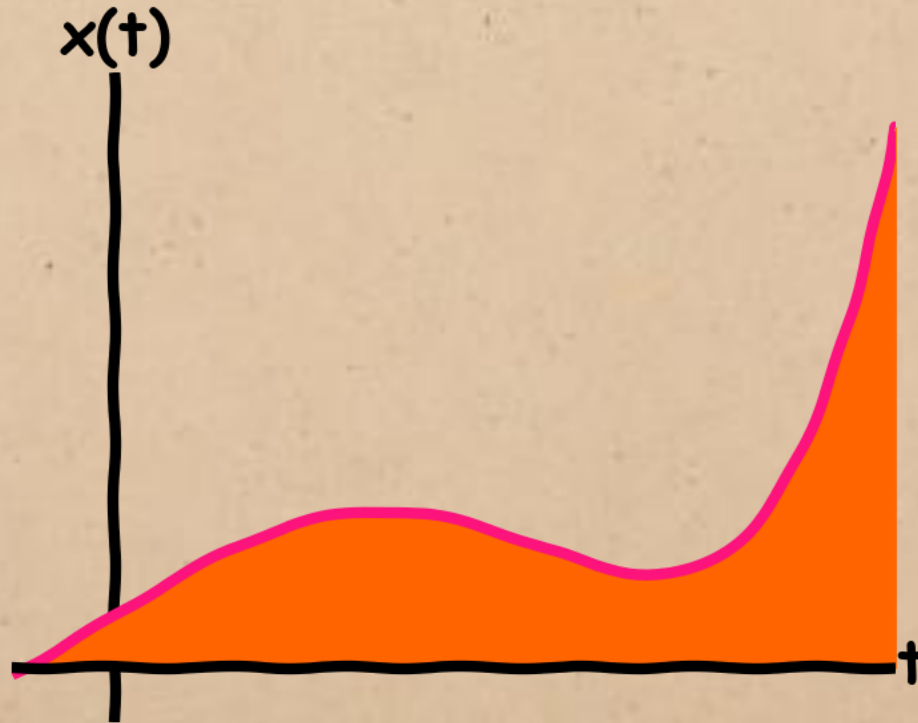
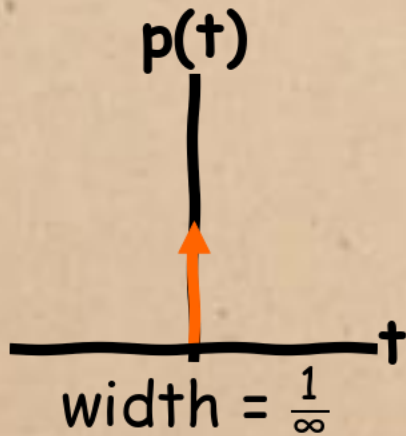


$$= \sum_{\lambda=-\infty}^{\infty} x\left(\frac{1}{8}\lambda\right) p\left(t - \frac{1}{8}\lambda\right)$$

LTI SYSTEMS

Narrower pulses spaced more closely together give a more accurate approximation.

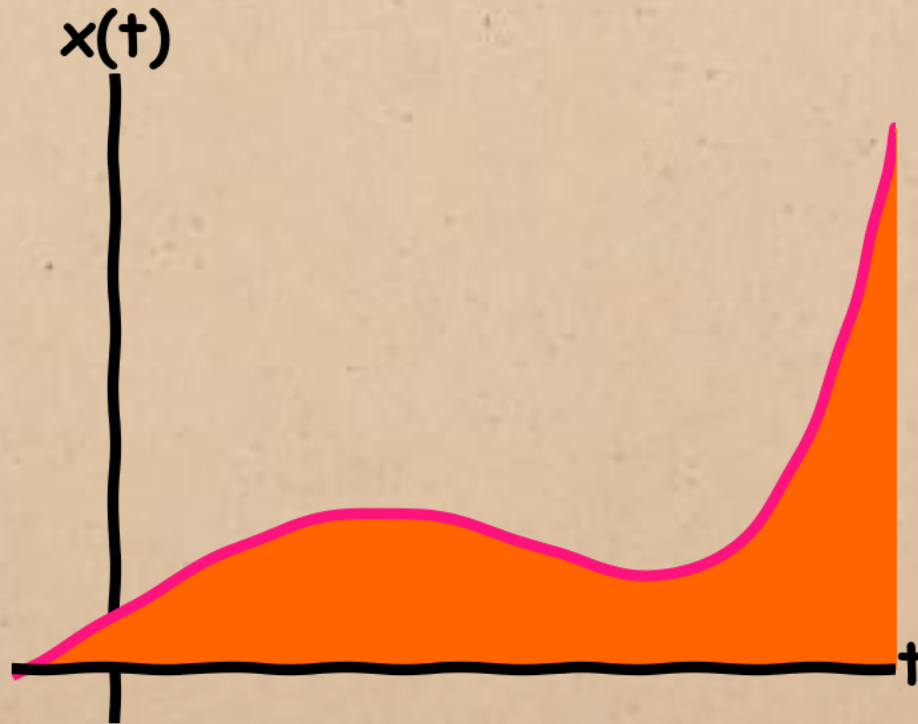
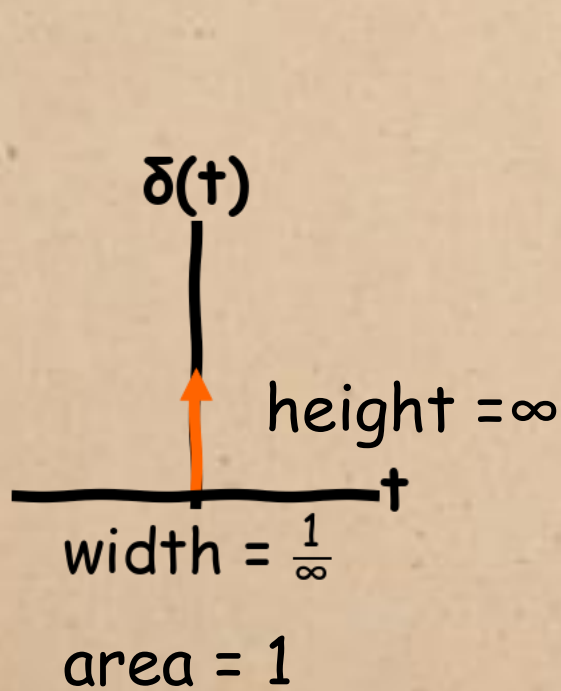
...so we'll get a perfect replica of the signal if we use an infinitesimally narrow pulse.



$$= \sum_{\lambda=-\infty}^{\infty} x\left(\frac{1}{\infty}\lambda\right) p\left(t-\frac{1}{\infty}\lambda\right)$$

IMPULSE

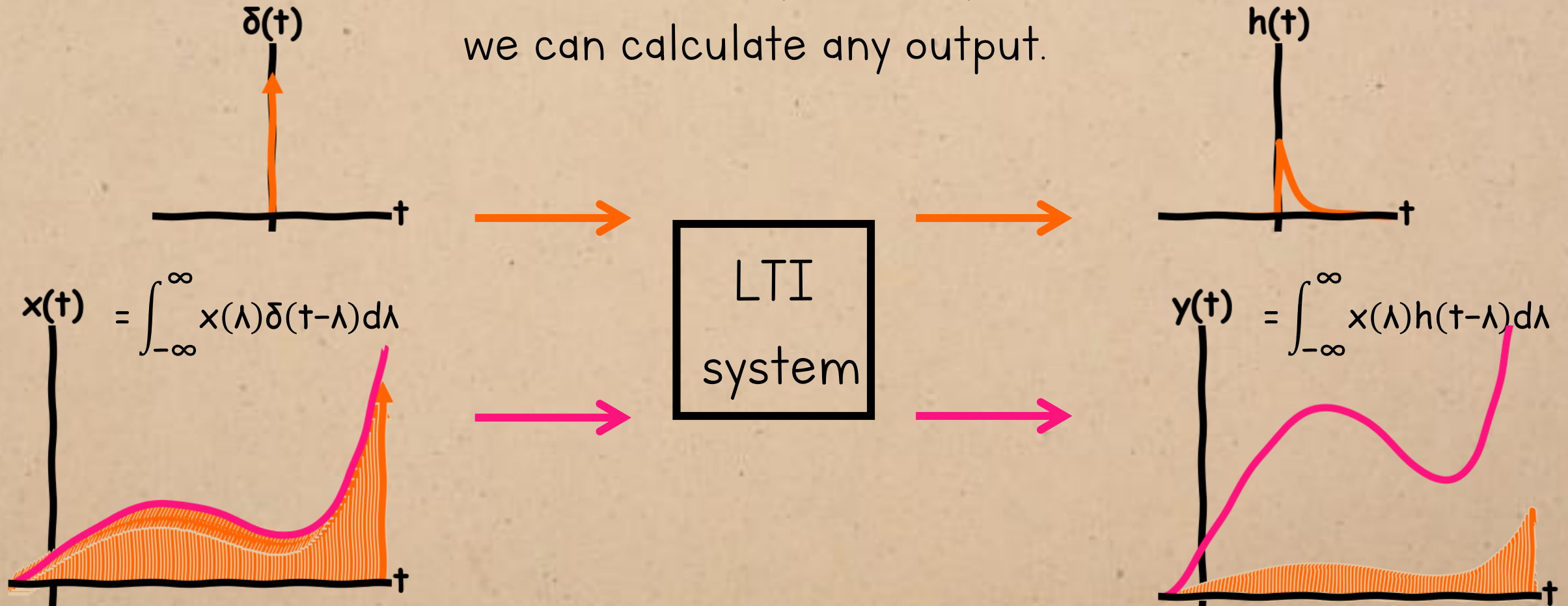
The Dirac delta (δ) or impulse function is an infinitesimally narrow pulse.



$$= \int_{-\infty}^{\infty} x(\lambda) \delta(t-\lambda) d\lambda$$

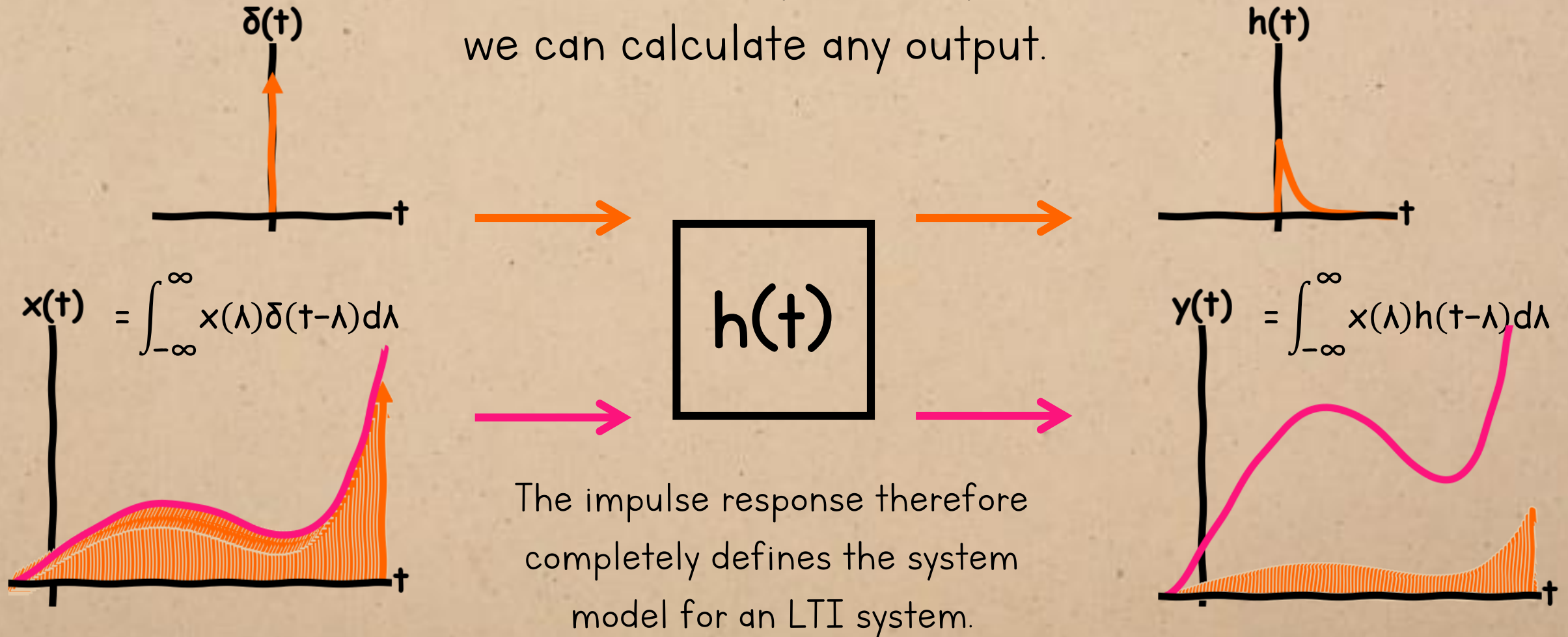
IMPULSE

...so if we know the **impulse response** $h(t)$,
we can calculate any output.



IMPULSE

...so if we know the **impulse response** $h(t)$,
we can calculate any output.



The operation we use to calculate the output from the input and impulse response is called **convolution**.

$$= \int_{-\infty}^{\infty} x(\lambda) h(t-\lambda) d\lambda$$

...But this is a computationally-intensive process, so we prefer to avoid it if possible.